

Arithmetic Inherits the Geometry of its Lattice

Euler, Pick, determinants, and the passage from Gaussian to Eisenstein integers

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Interactive companion: leelemacjames.github.io/euler-pick-lattice-learner.v3.html

Abstract

Four ideas — Euler’s topological invariant, Pick’s lattice formula, the determinant as area, and the passage from the square lattice to the triangular lattice of Eisenstein — are usually taught in separate courses. They are facets of one thing. This essay traces the single thread that connects them: the determinant of a lattice basis is the fundamental unit of area in that lattice, and every geometric formula expressed in that lattice inherits the determinant as its scaling factor.

A formula that survives deformation

Start with something that feels almost too clean to be true.

Take any convex polyhedron — a cube, a tetrahedron, a dodecahedron. Count its vertices V , edges E , and faces F . No matter what shape you choose,

$$V - E + F = 2.$$

Rescale it. Stretch it into something unrecognisable. The numbers V , E , F do not change. The formula holds.

This is not algebra. It is topology.

Lengths may distort. Angles may shear. But *connectivity* — which vertices are joined, which faces share an edge — survives every continuous deformation that does not tear or puncture the surface. The number 2 is the Euler characteristic χ of the sphere, and any convex polyhedron can be continuously inflated into a sphere, so every convex polyhedron inherits $\chi = 2$.

Topology preserves adjacency, not shape. That is why Euler's constant survives deformation.

The proof of Euler's formula makes this concrete. Project the polyhedron onto a plane — imagine shining a light through a transparent model and reading off the shadow. The three-dimensional solid becomes a planar graph, and the same count $V - E + F = 2$ holds for any planar graph. The dimension changed; the invariant did not.

This planar projection is the first hint of a pattern: the Euler formula will reappear, doing real work, when we turn to a completely different-looking problem about counting lattice points inside polygons.

Counting points to measure area

Draw a polygon on squared paper with all corners at grid intersections. Let I be the number of grid points strictly inside it and B the number lying exactly on the boundary. Pick's theorem says the area is

$$A = I + \frac{B}{2} - 1.$$

The formula looks miraculous. Area — a continuous, geometric quantity — emerges from a count of discrete points.

But the miracle dissolves when you see the proof, which turns out to be an application of the same Euler formula we just met.

The proof strategy

Subdivide the polygon into *primitive triangles*: triangles whose only lattice points are the three vertices themselves. Apply Euler's formula to the resulting planar graph. Let T be the number of triangles. With $V = I + B$ total lattice points and faces $F = T + 1$ (the triangles plus the single unbounded outer face), Euler gives $V - E + (T + 1) = 2$, so

$$T = E - V + 1.$$

Now count edges two ways. Each interior edge is shared by two triangles; each boundary edge borders only one. This gives $3T = 2E - B$, and therefore $E = (3T + B)/2$. Substituting back:

$$T = \frac{3T + B}{2} - (I + B) + 1 \implies T = 2I + B - 2.$$

The area formula now hinges on a single geometric fact about each primitive triangle.

Why every primitive triangle has area exactly $\frac{1}{2}$

We already likely know that a triangle has area $\frac{1}{2} \times \text{base} \times \text{height}$. Place one vertex of a triangle at the origin and call the other two vertices the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$. The base is $|\mathbf{v}_1|$ and the perpendicular height is $|\mathbf{v}_2| \sin \theta$, where θ is the angle between them. So

$$A_{\Delta} = \frac{1}{2} |\mathbf{v}_1| |\mathbf{v}_2| \sin \theta.$$

This expression has a name: it is the absolute value of the *determinant* of the matrix whose columns are $\mathbf{v}_1$ and $\mathbf{v}_2$:

$$A_{\Delta} = \frac{1}{2} |\det[\mathbf{v}_1, \mathbf{v}_2]| = \frac{1}{2} \left| \det \begin{bmatrix} v_{1x} & v_{2x} \\ v_{1y} & v_{2y} \end{bmatrix} \right|.$$

The determinant is not merely an algebraic device. It *is* the signed area of the parallelogram spanned by the two vectors.

Now, a triangle is *primitive* precisely when its edge vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ form a *basis* for the integer lattice $\mathbb{Z}^2$ — meaning every lattice point can be expressed as an integer combination of them. This is exactly the condition $|\det[\mathbf{v}_1, \mathbf{v}_2]| = 1$. Such matrices are called **unimodular**.

The pivotal fact: every primitive triangle has area $\frac{1}{2}$ because the unimodular condition forces $|\det| = 1$, making the triangle exactly half a unit parallelogram.

Since $A = T/2$ and $T = 2I + B - 2$, we obtain

$$A = I + \frac{B}{2} - 1. \quad \square$$

Pick's theorem is a theorem about determinants. The counting is Euler. The geometry is the determinant.

The area $\frac{1}{2}$ per primitive triangle is not a coincidence. It is the determinant of the basis of $\mathbb{Z}^2$, which equals 1, divided by 2.

The determinant as a geometric measure

Step back and ask what the determinant actually *means*.

Given a 2×2 matrix

$$M = \begin{bmatrix} a & c \\ b & d \end{bmatrix},$$

the determinant $\det(M) = ad - bc$ measures *oriented area scaling*: if you apply the transformation M to the unit square, the image has area $|\det(M)|$. Equivalently, the parallelogram spanned by the two column vectors has area $|\det(M)|$.

For a basis matrix — a matrix whose columns are the two vectors that generate a lattice — this has a concrete meaning: the determinant is the area of the *fundamental parallelogram*, the unit cell that tiles the plane.

On the integer square lattice $\mathbb{Z}^2$, the natural basis is

$$M_{\square} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \det(M_{\square}) = 1.$$

The fundamental parallelogram is the unit square. One unit of Euclidean area holds exactly one lattice cell. The lattice is *unimodular*.

This is why the classical form of Pick's theorem — $A = I + B/2 - 1$ without any prefactor — works on the square lattice. The determinant is 1, so it is invisible.

But change the lattice, and the determinant changes, and the formula changes with it.

Eigenvectors and the natural axes of a lattice

Before we change the lattice, it is worth pausing on eigenvectors, because they illuminate why the determinant has this area-measuring role.

A matrix M acts on vectors by stretching and rotating. A vector $\mathbf{v}$ is an *eigenvector* of M if the matrix does not rotate it — only scales it:

$$M\mathbf{v} = \lambda\mathbf{v}.$$

The scalar λ is the corresponding *eigenvalue*. Eigenvectors are the transformation's natural axes: directions it acts on most cleanly.

The key fact for us is this: the *product* of all eigenvalues equals the determinant. Since eigenvalues encode stretching factors along the natural axes, and area scaling is the product of all those stretching factors, the determinant measures total area scaling.

This is the geometric soul of the formula $\det(M) = \lambda_1 \lambda_2$.

For the square lattice $M = I$, every vector is an eigenvector with $\lambda = 1$. The lattice is maximally symmetric: all directions are equally natural. For the Eisenstein lattice, the eigenvalues deviate from 1, and the product $\lambda_1 \lambda_2 = \sqrt{3}/2$ measures the area of the tilted fundamental cell.

A second lattice: the Eisenstein integers

The integers $\mathbb{Z}$ generalise to the plane in more than one way.

The most familiar is the *Gaussian integers* $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$, where $i = \sqrt{-1}$. These are the complex numbers at integer coordinates, and they correspond exactly to the square lattice $\mathbb{Z}^2$ with basis vectors meeting at 90° .

But there is a second natural choice. Let

$$\omega = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

be a primitive cube root of unity, satisfying $\omega^3 = 1$ and $\omega^2 + \omega + 1 = 0$. The *Eisenstein integers*

$$\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$$

form a lattice in the complex plane, but now the two basis vectors 1 and ω meet at 60° rather than 90° . The resulting lattice is *triangular* — the densest packing of equal circles in the plane.

The basis matrix is

$$M_\omega = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} \end{bmatrix}, \quad \det(M_\omega) = 1 \cdot \frac{\sqrt{3}}{2} - 0 \cdot \left(-\frac{1}{2}\right) = \frac{\sqrt{3}}{2}.$$

The fundamental parallelogram of this lattice — the rhombus with angles 60° and 120° — has area $\sqrt{3}/2$, not 1. The lattice is *not* unimodular.

Comparing the two basis matrices:

Square lattice $\mathbb{Z}^2$

$$M_{\square} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\det = 1 \times 1 - 0 \times 0 = 1$$

Basis vectors at 90° . Unimodular.

$$\text{Primitive triangle area} = \frac{1}{2}.$$

Eisenstein lattice $\mathbb{Z}[\omega]$

$$M_{\omega} = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$\det = 1 \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$

Basis vectors at 60° . Not unimodular.

$$\text{Primitive triangle area} = \frac{\sqrt{3}}{4}.$$

Pick's theorem on the Eisenstein lattice

Now return to Pick's proof and ask: what changes on the Eisenstein lattice?

The combinatorial scaffold — triangulate, apply Euler, count edges — is identical. The logical structure does not care which lattice it is operating on. The *only* line that changes is the area per primitive triangle.

On the square lattice: primitive triangle area $= \frac{1}{2} |\det| = \frac{1}{2} \cdot 1 = \frac{1}{2}$.

On the Eisenstein lattice: primitive triangle area $= \frac{1}{2} |\det| = \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}$.

Since $A = T \times (\text{primitive triangle area})$, and $T = 2I + B - 2$ as before, the Eisenstein version of Pick's theorem is

$$A = \frac{\sqrt{3}}{2} \left(I + \frac{B}{2} - 1 \right).$$

The combinatorial formula $I + B/2 - 1$ is universal. The prefactor $\Delta = \det(M)$ encodes which lattice you are working in.

For a general lattice with fundamental domain area Δ :

$$A = \Delta \cdot \left(I + \frac{B}{2} - 1 \right).$$

The classical Pick formula is the special case $\Delta = 1$.

Pick's theorem is one formula, not two. The lattice determinant Δ is the single parameter distinguishing them. Arithmetic inherits the geometry of its lattice.

The cosine rule confirms the determinant

There is a satisfying elementary check.

The determinant of the Eisenstein basis matrix equals $|\mathbf{f}_1||\mathbf{f}_2|\sin\theta = 1 \cdot 1 \cdot \sin 60^\circ = \sqrt{3}/2$. That is the cross-product area formula.

But the same answer emerges from the cosine rule. The diagonal of the fundamental rhombus has length

$$d^2 = |\mathbf{f}_1|^2 + |\mathbf{f}_2|^2 - 2|\mathbf{f}_1||\mathbf{f}_2|\cos 60^\circ = 1 + 1 - 1 = 1,$$

so $d = 1$. The rhombus can be split into two isosceles triangles; the area of each is $\frac{1}{2} \sin 60^\circ = \frac{\sqrt{3}}{4}$, and the rhombus area is therefore $\frac{\sqrt{3}}{2}$.

Two completely different routes — the determinant, and the cosine rule — arrive at the same number. They are measuring the same fundamental cell from different directions. This kind of convergence is one of the signals of a genuine structural fact.

The thread in full

The ideas in this essay may seem to belong to different parts of mathematics. Euler's formula feels like topology. Pick's theorem feels like combinatorial geometry. Determinants feel like linear algebra. Gaussian and Eisenstein integers feel like number theory. They are not separate.

Euler's formula is a statement about planar connectivity. The proof of Pick's theorem recruits that connectivity result, then asks a geometric question: what is the area of a fundamental triangle? The answer is a determinant. The determinant is the area of the fundamental cell of the lattice. Change the lattice, and the determinant changes. And when the determinant changes, Pick's formula changes with it, in exactly the way you would predict if you understood the determinant as a measure of lattice density.

The passage from $\mathbb{Z}^2$ to $\mathbb{Z}[\omega]$ makes this concrete. The 90° basis of the Gaussian integers gives $\det = 1$. The 60° basis of the Eisenstein integers gives $\det = \sqrt{3}/2$. The Pick formula simply scales by that factor.

$$A = \Delta \cdot \left(I + \frac{B}{2} - 1 \right)$$

where $\Delta = \det(M_{\text{basis}})$ is the area of the fundamental domain.

This is what it means to say that arithmetic inherits the geometry of its lattice. The Eisenstein integers form a different grid and carry a different measure of area — a different ruler — and every formula you derive on that lattice will reflect that ruler.

The determinant is that ruler.

A note on what this opens

This essay stops before the deeper water, but it is worth pointing toward it.

The Eisenstein integers $\mathbb{Z}[\omega]$ have a special property: they form a *unique factorisation domain*. Every Eisenstein integer factors into Eisenstein primes in essentially one way, just as every ordinary integer factors into primes. The 60° geometry of the triangular lattice is a result of the basis angle being a geometric reflection of the arithmetic structure of $\mathbb{Z}[\omega]$.

The class number of the number field $\mathbb{Q}(\omega)$ is 1, which is the algebraic statement that unique factorisation holds. The lattice determinant $\sqrt{3}/2$ is related to the *discriminant* of that number field. These are not separate facts; they are the same fact expressed in different languages.

The Gaussian integers $\mathbb{Z}[i]$ have the same property. So do the rings associated with several other quadratic fields. The lattices for which this works most cleanly are related to the Heegner numbers — a story connecting unique factorisation, class numbers, and the remarkable near-integers produced by expressions such as $e^{\pi\sqrt{163}}$.

The thread that began with counting corners of a polyhedron leads, eventually, to some of the deepest structures in number theory.

Interactive companion learner (drag polygons, shear the lattice, watch the determinant and Pick formula update live):

leelemacjames.github.io/euler-pick-lattice-learner_v3.html